

Tanuj Sharma

Senior Software Engineer – Backend | Credit Cards, Lending & Payments Infrastructure

Gurugram, India | +91 80948 45731 | tanujsharma1412@gmail.com | linkedin.com/in/tanuj-sharma1412

SUMMARY

Backend engineer with 5+ years building credit card and lending systems (CMS, Credit Line on UPI, EMI engines, disputes, double-entry ledgers) with Java, Spring Boot, Kafka, and PostgreSQL on AWS.

SKILLS

Fintech Domain: Card Issuance, CMS, LMS, Credit Line on UPI, RuPay, EMI, Disputes, Double-Entry Ledger

Standards & Compliance: ISO 8583, NPCI Settlement Reconciliation, PCI-DSS (audit), Idempotent APIs

Languages & Frameworks: Java 11, Spring Boot, Golang, Python, SQL, REST APIs, Webhooks, JUnit, Mockito

Data & Messaging: PostgreSQL, MySQL, Apache Kafka (Transactions, Outbox, DLQ), Redis

Cloud & Observability: AWS (ECS, RDS Aurora, Lambda, SQS, S3), Docker, Kubernetes, CI/CD, Prometheus, Grafana

EXPERIENCE

Vegapay

Senior Software Engineer – Backend

Mar 2023 – Present

Gurugram, India

B2B credit card and lending infrastructure for banks and fintechs. Joined as the 11th employee; company now 135+.

- Designed and built the core Credit Card Management System (CMS) and the Credit Line on UPI platform before the team scaled; it now handles **1M+ transactions/day** and **INR 1,344 Cr+** lifetime volume.
- Owned the Unity × BharatPe RuPay credit card backend end-to-end: issuance, ISO 8583 authorization (including international transactions), limits, EMI engine, and double-entry ledger.
- Technical point of contact for Unity Bank and BharatPe through the card go-live; also integrated Credit Line on UPI programs for SSFB × Paytm and YBL × BharatPe.
- Run 12 production microservices at **99.99% availability** and a **99.99% transaction success rate**, with P99 authorization latency under **150ms**.
- Built daily reconciliation of NPCI settlement files (**1M+ records/day**) against the double-entry ledger.
- Banks were escalating duplicate and silently dropped transactions: made APIs and Kafka consumers idempotent (DB unique constraints) and added retries, DLQ, and alerts, cutting escalations **70%**.
- Designed and built the dispute management service from scratch, modelling each dispute as a state machine with validation and reconciliation from raise to closure.
- Designed and built daily book closure as parallel workflows across services with optimized DB access, producing **800K+ user bills within an hour**.
- Fixed an EMI ledger double-posting race (sync and async Kafka write paths) seen only under production latency: moved EMI to one sync writer, reconciled affected entries, and set a single-writer rule for ledger flows.
- Mentor 4–5 engineers; guided the design and reviewed the delinquency and collections module they built (event-driven bucket movements, **INR 29 Cr+** collected).

BharatPe

Software Developer

Jul 2021 – Mar 2023

Gurugram, India

- Built BharatPe Card and PostPe card lifecycle features (1M+ active accounts): onboarding, activation, spend limits.
- Integrated card platforms Happay and Zeta so card numbers were not stored in BharatPe systems.
- Optimized onboarding and authorization queries and indexes, cutting latency from **0.9–1.5s to under 400ms**.
- Customers were stalling between onboarding stages; debugged the failures and restructured the flow logic, getting **45%** of stuck customers fully onboarded.

Grab

Software Engineer Intern

Feb 2021 – Jul 2021

Remote / Bangalore, India

- Built a concurrent Golang diagnostic service that detects failovers and connection-pool exhaustion across MySQL, PostgreSQL, and MSSQL, cutting developer MTTR by **80%**.

EDUCATION

B.Tech, Computer Science & Engineering, Poornima College of Engineering, Jaipur

2017 – 2021